

TwinPulse

ML-Based Digital Twin Framework for AI-Driven Asset Prognosis

Group 6 | Project Team

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The Problem

- Reactive maintenance relies on failure events or static thresholds
- Unplanned downtime drives cost escalation and production loss
- Zero visibility into asset lifecycle or degradation rates



The Objective

- Predict degradation 5–30 days in advance
- Convert raw sensor telemetry into actionable operational intelligence
- Shift from reactive repairs to condition-based maintenance

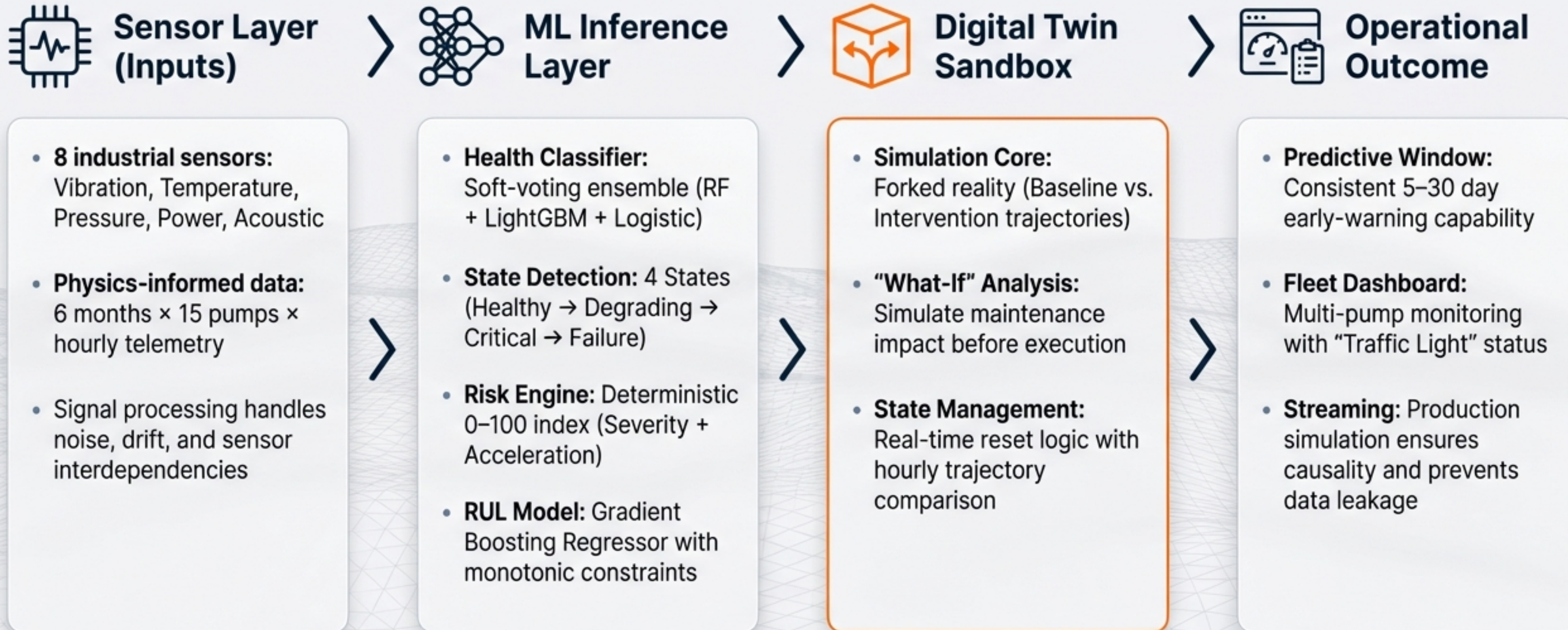


What We Achieved

- 4-stage health lifecycle modelling (Healthy → Failure)
- Continuous 0–100 deterministic risk scoring
- Monotonic RUL forecasting with physical constraints
- Streaming-ready Digital Twin with real-time feedback

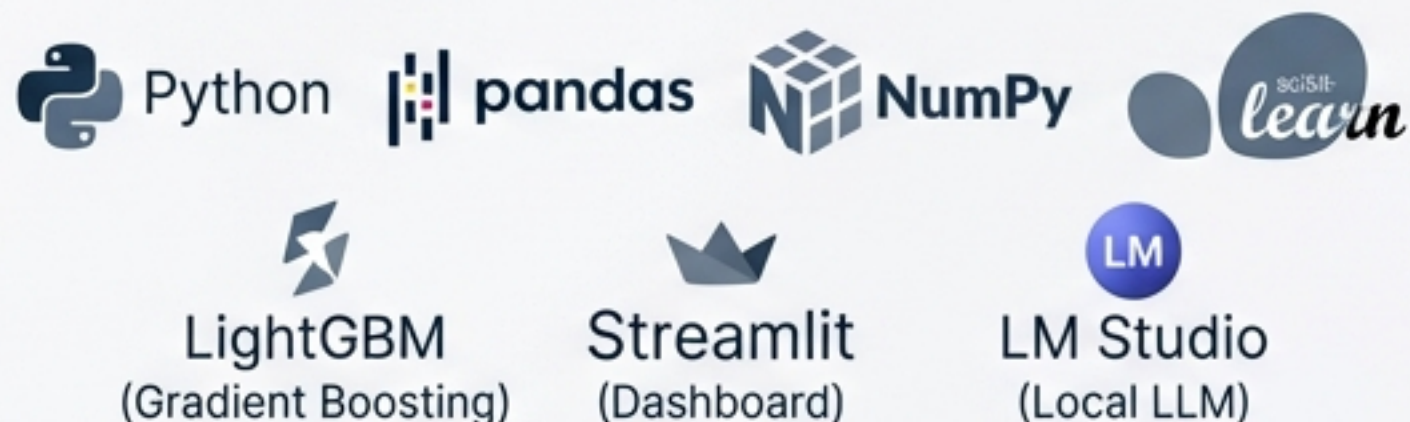
Target Users: Plant Operations Heads | Maintenance Managers | Reliability Engineers | Asset Performance Teams

System Architecture: From Sensor Data to Decision Support



Technology, Differentiation & Product Value

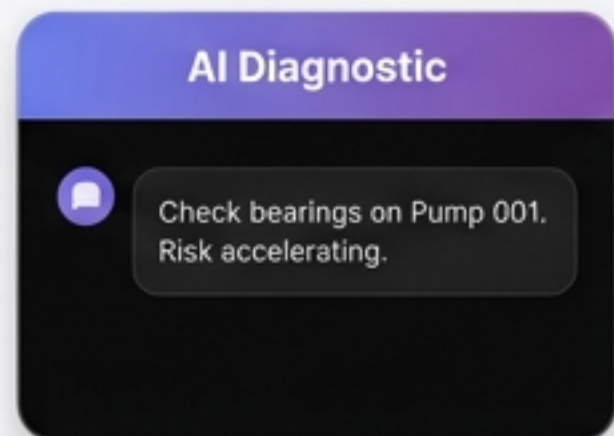
Tech Stack



Differentiation

- ✓ **End-to-End Architecture:** Unified pipeline, not just isolated models
- ✓ **Hybrid Intelligence:** Combines ML probability with deterministic risk rules
- ✓ **Physics-Constrained:** Monotonic health latching prevents unrealistic forecasts
- ✓ **Interactive Sandbox:** Unique capability to simulate "What-If" repair scenarios

AI Diagnostic Agent



- **Natural Language:** Translates sensor spikes into plain-English insights
- **Explainability:** Provides risk-based maintenance recommendations
- **Context:** Analyzing sensor-level root causes for the operator

Business Value Proposition

- ↗ **Downtime Reduction:** Eliminate unplanned outages via early detection
- ↗ **Resource Optimization:** Prioritize maintenance based on risk, not calendar
- ↗ **Fleet Visibility:** Instantly identify the "Critical Few" assets 
- ↗ **Cost Control:** Shift emergency repairs to planned, lower-cost interventions

TwinPulse transforms raw sensor data into predictive operational intelligence.